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Spatial Aware Peak Ranking in Mass Spectrometry Imaging

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***Abstract*—Mass spectrometry imaging (MSI) records thousands of mass-to-charge (m/z) channels at every pixel of a tissue section, so analysis usually begins by selecting a few informative peaks. Learned methods such as msiPL rank peaks from an unsupervised model, and S^{3}PL adds spatial information through 3D convolution. We ask whether local neighbourhood context improves learned peak ranking and where in the pipeline it matters. Variational autoencoders with and without neighbourhood input were segmented with a Gaussian mixture model. Peaks were ranked either by Integrated Gradients (IG) through the model or by correlating each ion image with the segment posterior maps. On 24 sections from three public datasets, at matched peak budgets and under predeclared decision rules, correlation outperformed IG in 140 of 144 paired comparisons across three seeds, a difference that was statistically supported for every dataset and model. Neighbourhood input gave no consistent gain, whereas spatially aware ranking statistics were the strongest label-free rankings. Spatial information therefore helps at the ranking stage rather than at the encoder input.**

***Index Terms*—mass spectrometry imaging, peak selection, variational autoencoder, spatial context, Integrated Gradients, reproducibility**

I. INTRODUCTION

Mass spectrometry imaging (MSI) records a full mass spectrum at every pixel of a tissue section, so the spatial distribution of many molecules can be studied at once without choosing targets in advance [Caprioli et al. 1997; Schwamborn and Caprioli 2010; Buchberger et al. 2018]. The resulting data are large and high-dimensional. The collections used here contain between about a thousand and several thousand pixels per section, and between 1,481 and 85,062 mass-to-charge (m/z) bins per spectrum. Most analyses therefore begin by selecting a small set of informative peaks.

Classical *peak picking* applies smoothing, normalisation and signal-to-noise thresholds to each spectrum on its own [Gibb and Strimmer 2012; Deininger et al. 2011]. It depends on manually set thresholds, can discard weak but informative ions [Lieb et al. 2020] and ignores *where* in the tissue a signal occurs. *Peak learning* instead treats selection as an unsupervised representation-learning problem: a model is fitted to the spectra of a section and m/z bins are ranked by how the model uses them [Li et al. 2022; Abdelmoula et al. 2021]. msiPL trains a variational autoencoder (VAE) on individual spectra and reads peaks from its weights [Abdelmoula et al. 2021]. S^{3}PL adds spatial information by convolving over small tissue patches and reconstructing the whole neighbourhood, and reports large gains over msiPL [Weigand et al. 2026a].

S^{3}PL changes the architecture, the objective, the normalisation and the selection rule at the same time, so it is not clear which ingredient produces its gain. This raises two questions. Can cheaper local context, such as the mean spectrum of a pixel's neighbours, give a lightweight VAE a similar benefit? And once a model has been fitted, which part of the pipeline actually determines peak quality: the representation, the segmentation built on it or the method used to rank peaks? We study these questions with *Spatial-msiPL*, an msiPL-style VAE that can encode a spectrum together with its neighbourhood. Its latent space is segmented with a Gaussian mixture model (GMM), and peaks are ranked either by Integrated Gradients (IG) [Sundararajan et al. 2017] through the model or by correlating each ion image with the GMM posterior maps. Specifically, we ask:

RQ1 Does the VAE pipeline select better peaks than msiPL when both select the same number of bins?

RQ2 Which stage of the pipeline determines peak quality: the attribution method or the segmentation, and does the segmentation need a nonlinear encoder?

RQ3 Where does spatial information help: as input to the encoder, in the segmentation or in the ranking statistic?

RQ4 Do the published msiPL and S^{3}PL results reproduce, and how sensitive are they to the training seed?

RQ1 and RQ4 establish that comparisons are fair and that the baselines are what they are reported to be. RQ2 and RQ3 locate the source of any gain, which a single headline score against a baseline cannot do.

The evaluation was designed to be fair rather than favourable. Every method selected the same number of bins per section, every decision rule was written down before the corresponding results existed, the main comparisons were repeated over three training seeds, and reproduced msiPL and S^{3}PL baselines and simple model-free rankings (mean intensity, variance and Moran's I) were scored in the same way. The main findings are as follows.

- **Ranking.** With the segmentation held fixed, correlating each ion image with the GMM posterior maps outperformed IG in 140 of 144 paired comparisons across three training seeds. The difference was statistically supported for every dataset and model.
- **Spatial information.** Neighbourhood input to the encoder gave no consistent, spatially specific gain. Spatially aware ranking statistics did: correlation with the segmentation maps and Moran's I were the strongest label-free rankings. Peak ranking benefits from being spatially aware, which gives this report its title.
- **Encoder.** A linear five-component PCA segmentation performed comparably to the VAE, so the nonlinear encoder was not shown to be necessary.
- **Reproducibility.** msiPL reproduced. The published S^{3}PL glioblastoma score lay outside the distribution of ten reproduced seeds, and seeds changed S^{3}PL mainly through weight initialisation, which switched individual sections between distinct solutions.

The contributions of this report are:

1) a predeclared, matched-budget evaluation of learned MSI peak ranking on three public datasets, with reproduced baselines, model-free references and three training seeds;
2) a ranking ablation showing that segment-then-correlate outperforms gradient attribution in this pipeline, with paired statistics;
3) controlled tests of spatial context at the encoder, segmentation and ranking stages, which bound where spatial information helps;
4) a statistical account of the S^{3}PL reproduction gap and its seed sensitivity.

Section II reviews MSI peak selection and the methods compared. Section III describes the data, models, evaluation, statistics and computation. Section IV presents the results, section V interprets them and discusses limitations, and section VI concludes.

II. Background and Related Work

A. MSI data and preprocessing

An MSI experiment rasters a tissue section and records a spectrum at each location, giving a data cube with two spatial dimensions and one m/z dimension [Caprioli et al. 1997; Buchberger et al. 2018]. Spectra vary in total intensity between pixels for technical reasons, so they are usually normalised, most often by total ion current (TIC), before comparison [Deininger et al. 2011]. Spectra may be kept as fine m/z bins or reduced to centroided peaks. Because neighbouring bins of one molecular peak are strongly correlated, any evaluation that counts bins can reward several bins of the same peak. This motivates the peak-level scoring used in this report (section III-C). Tools such as Cardinal [Bemis et al. 2015] and M^{2}aia [Cordes et al. 2021] provide standard data handling. For our work, the key consequence is that every method must receive the same input and be scored at the same resolution, or differences in preprocessing masquerade as differences in method.

B. Peak selection

Classical pipelines such as MALDIquant detect local maxima above a noise threshold in each spectrum [Gibb and Strimmer 2012]. Machine-learning peak pickers learn what a peak looks like from annotated spectra [Li et al. 2022]. Both operate spectrum by spectrum and do not use the spatial arrangement of the tissue. Learned representation methods instead rank m/z bins by how a model trained on the whole section uses them. This report compares such learned rankings with each other and with simple model-free rankings, under one shared evaluation.

C. Learned peak selection: msiPL and S^{3}PL

msiPL trains a fully connected VAE [Kingma and Welling 2014] on individual spectra, clusters the low-dimensional latent space and reads peaks from the network weights [Abdelmoula et al. 2021]. Its companion work, massNet, provides the glioblastoma (GBM) dataset used here [Abdelmoula et al. 2022]. S^{3}PL learns a 3D convolution over a spectral patch around each pixel, applies an attention step to the centre spectrum, reconstructs the full normalised patch and selects the bins that receive the most attention across pixels [Weigand et al. 2026a]. S^{3}PL also introduced the mask-based spatial-correlation F1 score (mSCF1) used to judge peak lists against expert annotations, and released the peak budgets and protocols for GBM, colorectal adenocarcinoma (CAC) and renal cell carcinoma (RCC). It reported 0.496, 0.601 and 0.470 mSCF1 on these collections, against 0.403, 0.431 and 0.159 for msiPL. Related representation-learning work in MSI includes deep models of 3D data [Inglese et al. 2017], low-dimensional ion-image embeddings [Guo et al. 2024] and IonMorphNet [Weigand et al. 2026b]. S^{3}PL's result motivates our spatial question. Its convolution, objective, normalisation and selector all differ from msiPL, so it is not clear which ingredient produces the gain, or whether a much smaller amount of spatial context would suffice.

D. Representation quality versus downstream use

A representation that reconstructs the data well is not necessarily the one that serves a later task best. Mazarura, Jarvis and Bau compared autoencoders, PCA and NMF on MSI data and found that reconstruction error, cluster quality and downstream classification ranked the models differently [Mazarura et al. 2025]. We test the same distinction for peak ranking (section IV-E).

E. Attribution and segment-then-correlate

Integrated Gradients attributes a model output to input features by integrating gradients along a path from a baseline input to the actual input [Sundararajan et al. 2017]. Applied to a GMM posterior computed from a VAE latent space, it ranks m/z bins by how strongly they move a pixel towards a tissue segment, through the full nonlinear encoder. An older and simpler route is to segment the tissue first and then correlate each ion image with the segments [Alexandrov et al. 2010; Alexandrov and Kobarg 2011]. Our pipeline contains both routes on the same segmentation, which allows a direct comparison.

TABLE I
DATASETS. SECTIONS ARE THE UNITS OF ANALYSIS; THE INDEPENDENT UNITS BEHIND THEM DIFFER BY COLLECTION.

	GBM	CAC	RCC
Sections	8	8	8
Independent units	4 PDX lines	1 tumour	8 subjects
m/z bins	85,062	1,481	10,200
Pixels per section	2,071–4,524	3,243–4,761	972–3,564
Mask classes (GMM K)	2	3	3
Ionisation	MALDI	DESI	DESI

F. Spatial statistics and spatially aware methods

Ion images of biologically meaningful molecules tend to be spatially structured rather than noisy. Moran's I measures spatial autocorrelation [Moran 1950], and spatial autocorrelation and spatial-chaos measures have been used to judge the informativeness of MSI ion images [Alexandrov and Bartels 2013; Cassese et al. 2016]. Spatially aware segmentation smooths features using neighbourhood weights before clustering [Alexandrov and Kobarg 2011]. Graph-based models of spatial omics and MSI propagate information between neighbouring locations. These are discussed as future work in section VI. Our study places spatial information at three points (the encoder input, the segmentation and the ranking statistic) and tests each one under the same evaluation.

G. Gap addressed

Published learned peak-selection methods are usually evaluated against one or two baselines, often with different peak counts, single training runs and no model-free references. It has not been tested under a common budget whether their advantages come from spatial input, from the learned representation, or from the ranking step, nor how stable those advantages are across training seeds. This report addresses that gap for a VAE-based pipeline, with reproduced msiPL and S^{3}PL baselines.

III. METHODOLOGY

A. Data

Three public MSI collections were used (table I and figure 1). They are the collections on which S^{3}PL was published [Weigand et al. 2026a], so both baselines could be reproduced under their own protocols and then compared with our methods under a common one.

GBM is the massNet patient-derived xenograft (PDX) glioblastoma collection, acquired by MALDI on a 9.4 T FT-ICR instrument and deposited in the Metabolomics Workbench (project PR001292, study ST002045) [Abdelmoula et al. 2022]. It contains eight sections from four PDX lines, two per line, with pathologist annotations of tumour and normal tissue. Sections from the same line are not independent, so GBM results are also summarised per line. **CAC** is a DESI collection of 52 serial sections from one human colorectal adenocarcinoma (MetaboLights MTBLS415) [Inglese et al. 2017]. We used the eight sections and three-class masks released with S^{3}PL [Weigand et al. 2026a], so CAC comparisons are within one specimen. **RCC** is the DESI renal cell carcinoma collection of Dill et al. [2010], distributed with Cardinal [Bemis et al. 2015]. It contains eight runs, each a matched pair of cancer and normal tissue from one subject. Pixels outside the annotated tissue have no label; this unannotated region covers 45–78% of each run. The dataset's own workflow treats it as background. We kept it as a third class in the primary analysis and report a cancer/normal-only sensitivity analysis (section IV-G). Measured coordinates and masks were audited for alignment in every section.

Four sections (GBM108_positive, GBM22_2, 160TopL and 40TopL) were used for development and tuning. The method was frozen on 16 September, before the comparisons reported here were run. RCC was not used during development and serves as a held-out confirmation collection.

B. Pipeline

For each section, a model is fitted to that section's spectra without labels, and the selected peaks are then scored against that section's expert masks. The design is therefore unsupervised and transductive. The masks never enter model fitting or ranking. They do, however, set two configuration values, the number of GMM components and the peak budget, both disclosed below.

VAE. Spectra are normalised by total ion current (TIC). Following msiPL, a fully connected VAE with one hidden layer of 512 units and a 5-dimensional latent space is trained for 100 epochs with Adam (learning rate 10^{-3}, batch size 128). The loss is msiPL's categorical cross-entropy reconstruction loss plus a KL term with weight 1 [Abdelmoula et al. 2021; Kingma and Welling 2014]. Two core arms are compared: *centre-only*, which encodes one spectrum, and *uniform*, which concatenates the spectrum with the mean of its up to eight measured neighbours in a 3×3 window. Both reconstruct the centre spectrum only. For GBM the two arms have 87.2M and 130.8M parameters. Further variants, trained with seed 1, were:

- learned single-head attention over the neighbours [Vaswani et al. 2017], which adds about 0.5% of parameters;
- uniform windows of 5×5, 7×7 and 9×9;
- zeroed neighbours and neighbours shuffled from distant pixels;
- centre masking, in which the centre spectrum is hidden for half of the training samples so that the neighbours become necessary.

Segmentation. A GMM with full covariance is fitted to the standardised latent means. Its number of components K equals the number of annotated classes (table I). This is a class-informed choice; the Bayesian information criterion (BIC) [Schwarz 1978] did not support it (section IV-E). The pipeline is therefore label-free in fitting and ranking but not fully autonomous.

Rankings. Every ranking orders all m/z bins, and the top K_{bin} bins are scored.

- *Integrated Gradients* integrates the gradient of the assigned GMM component's posterior with respect to the input spectrum [Sundararajan et al. 2017]. The path runs from the section's mean spectrum to each of 12 sampled pixels per

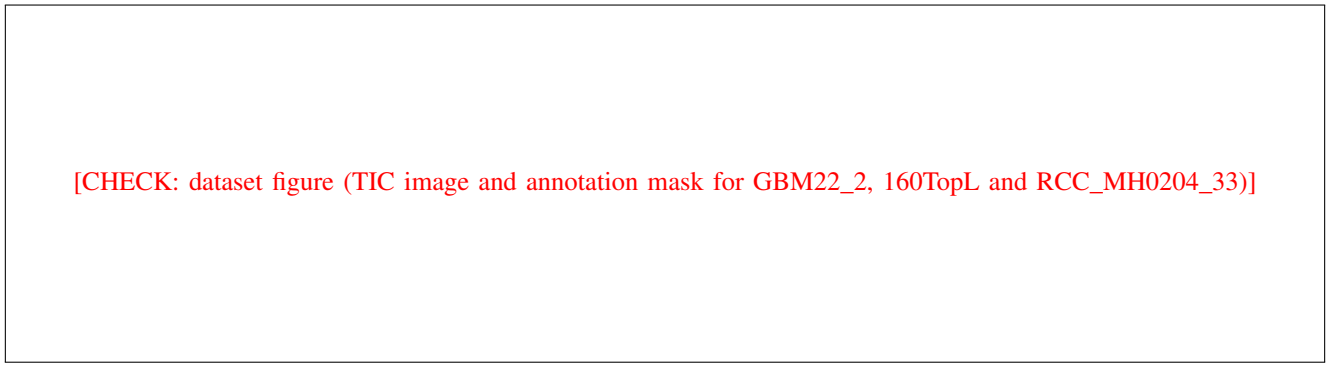


Fig. 1. One representative section per collection. Top: total ion current image. Bottom: expert annotation. Unmeasured locations are blank; on RCC the unannotated region is grey.

component, with 64 trapezoidal steps over both the centre and the context inputs. Bins are ranked by mean absolute attribution, balanced across components. Each run had to pass IG's completeness check (attributions summing to the change in output). Where it failed (4 of 144 runs), the run was recomputed from its saved model with 128 or 256 steps. In four control runs, doubling the steps changed mSCF1 by at most 0.0003.

- *Posterior correlation* ranks bins by the absolute Pearson correlation between each bin's ion image and each component's posterior map, balanced across components in the same way. It uses the same GMM as IG and no gradients; this is the segment-then-correlate approach [Alexandrov and Kobarg 2011].
- *Model-free rankings:* mean intensity, variance and global Moran's I over measured Moore-neighbour edges [Moran 1950].
- *References:* a random ranking, and a supervised oracle that ranks bins by correlation with the expert masks. The oracle is a ceiling only.

Controls on the representation and segmentation. A *PCA control* replaces the VAE latent space with a 5-component PCA while keeping the GMM and posterior correlation, which isolates the nonlinear encoder. A *smoothed segmentation* smooths the latent means with structure-adaptive neighbourhood weights before the GMM [Alexandrov and Kobarg 2011], which places spatial information at the segmentation step instead of the encoder input.

Baselines. Legacy msiPL was reproduced from its released code, with its peak-selection threshold tuned towards the published peak counts as in the original procedure [Abdelmoula et al. 2021]. S^{3}PL was run from the public repository (v1.0.0) with the released settings: 10 epochs, Adam (learning rate 10^{-2}), batch size 16, and patch size and peaks per patch of $p = 3$, $z = 256$ (GBM), $p = 9$, $z = 256$ (CAC) and $p = 5$, $z = 1024$ (RCC) [Weigand et al. 2026a]. The released code normalises each patch by its spatial maximum, whereas the paper describes TIC normalisation, so both were tested (section IV-F).

C. Evaluation

mSCF1. For each class mask and each threshold $t \in \{0.3, 0.4, 0.5, 0.6\}$, a reference set of bins is formed from the bins whose ion images correlate with the mask, using the threshold-derived rank cutoff of the released S^{3}PL evaluator. The F1 score between the selected bins and the union of the class reference sets is computed at each threshold, and mSCF1 is the mean over the four thresholds [Weigand et al. 2026a]. It measures spatial agreement with expert annotation, not chemical identity.

Matched budgets. All methods select the same number of bins, K_{bin}, per section. For GBM and CAC this is the tuned legacy msiPL peak count. For RCC it is the released S^{3}PL target, which approximately equals the number of bins correlating with the masks at $t = 0.5$. Budgets of $0.5\times$ to $2\times K_{\text{bin}}$ were checked as a sensitivity analysis. [CHECK: msiPL budget re-tune: one sentence on how msiPL was matched at 0.5–2× and on RCC (item 2)]

Peak-level scoring (P1). Neighbouring bins of one molecular peak would otherwise each count as a true positive. P1 therefore collapses bins within one basin of the section's mean spectrum into a single peak group, and partitions were audited and approved before any score was computed. On CAC and RCC, P1 groups are singletons or near-singletons, so P1 acts there as a second budget check rather than independent evidence.

D. Experimental design

Every comparison in section IV has a decision rule that was written down before its outputs existed, in an append-only protocol that also records deviations and execution logs. A difference *helps* if its mean is at least $+0.02$ mSCF1 and it is positive in at least 6 of 8 sections. Two methods are *equivalent* if the mean difference lies within ± 0.02. Anything else is *mixed*. These practical labels are quoted exactly in the results. All arms were trained with seed 1, and the two core arms were retrained with seeds 2 and 3 on all 24 sections. Collection-level seed results are reported as mean $\pm$ SD across seeds of the seed-level collection means.

E. Statistical analysis

The practical labels are accompanied by paired non-parametric statistics over sections: the exact two-sided Wilcoxon signed-rank test [Wilcoxon 1945], the Hodges–Lehmann (HL) median paired difference with its exact 95%

confidence interval [Hodges and Lehmann 1963], the matched-pairs rank-biserial correlation r and the sign count. Where three seeds exist, seeds are averaged within each section first, so the unit is the section rather than the run. Because every pair of methods is scored on the same sections, paired tests are used throughout.

The 38 rule-labelled comparisons were grouped into five families by question: ranking (6 tests), versus legacy msiPL (4), encoder context and variants (17), segmentation (9) and S^{3}PL (2). Holm correction [Holm 1979] was applied within each family. With $n = 8$ the smallest attainable exact p-value is $2/2^8 = 0.0078$, so Holm significance is attainable only in families of at most six tests, and effect sizes and intervals carry the main weight. A comparison is called *statistically supported* only if its Holm-adjusted $p < 0.05$ *and* its practical label points the same way. GBM is also summarised per PDX line ($n = 4$), outside the Holm families. Mixed-effects models with random intercepts for section and seed are reported descriptively.

F. Implementation and computation

The pipeline was implemented in Python with PyTorch and scikit-learn [Pedregosa et al. 2011]. Training ran on the University's MSCluster under Slurm, on RTX 3060 and RTX 3090 GPUs, and ranking and scoring ran on CPU nodes.

Caching. The original data loader re-read each batch from HDF5 on shared storage, and a 100-epoch GBM run took 11.3 h. Profiling showed that preparing a batch took 24.6 s, while the forward and backward passes took about 0.02 s. Reading each section into memory once (about 13 s) reduced the run to 23 min, a 30× speed-up, with identical loss histories and final weights. The likely cause is many small reads over network storage, which was not measured separately.

Reproducibility checks. Before saved runs were reused, same-seed reruns were checked for exact equality of losses, scores and selected bins. Exact equality of rankings depends on the CPU model, because floating-point summation order differs between processors, so timing runs were matched to the processor that produced each saved ranking. The code, protocol, result summaries and figure scripts are available at https://github.com/2653934/Spatial-msiPL[CHECK: release tag and commit].

Use of AI tools. [CHECK: Zayd to confirm wording] Generative AI assistants were used throughout the project, as declared on the final page. Claude (Anthropic) was used to help plan experiments, draft decision rules and review results against the saved outputs. Claude Code was used to write, test and run analysis code and cluster jobs, and OpenAI Codex was used to review code changes before they were committed. AI assistance was also used to draft and edit the text of this report. Every experiment, decision rule, cluster job and result was approved by the author, and every reported number was checked against a saved result file.

IV. Results

Unless stated otherwise, scores are mSCF1 at the matched bin budget K_{bin} and use training seed 1. Differences are paired by section. Practical labels are quoted exactly, and the statistics of section III-E are reported beside them.

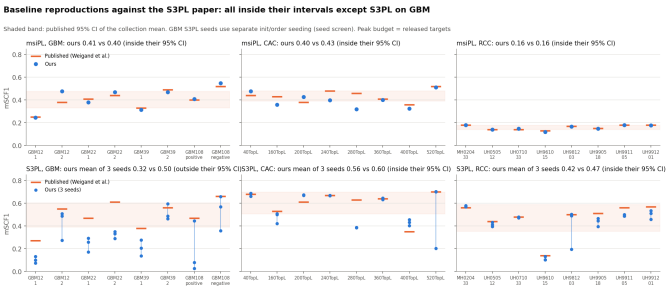


Fig. 2. Reproduced against published collection means for msiPL and S^{3}PL, each under its own protocol and evaluator. msiPL reproduces on all three collections; S^{3}PL reproduces on CAC and RCC but not on GBM.

A. Baseline reproductions

Figure 2 compares the reproduced baselines with the published values, each under its own protocol and evaluator.

- **msiPL** reproduced closely on all three collections: GBM 0.414 against 0.403 published, CAC 0.402 against 0.431, and RCC 0.157 against 0.159.
- **S^{3}PL on CAC and RCC** was close: 0.592 against 0.601 on CAC at seed 1, and 0.41–0.44 against 0.47 on RCC over three seeds.
- **S^{3}PL on GBM was not reproduced:** 0.32 against 0.496 published. We used the public repository with the released settings. The paper's supplementary tables were not publicly available, so its exact per-dataset settings could not be checked. The gap is analysed in section IV-F.

B. Ranking by correlation outperforms IG (RQ1, RQ2)

Table II gives all rankings scored on the same segmentation and budget.

Against legacy msiPL (RQ1). Both VAE–IG arms beat reproduced msiPL in every GBM and CAC section: 0.518 and 0.511 against 0.414 on GBM, and 0.528 and 0.560 against 0.402 on CAC. This is statistically supported in all four comparisons (HL $+0.09$ to $+0.16$, $r = 1$, Holm $p = 0.031$). [CHECK: budget-stability clause across 0.5–2 × K_{bin} (item 2)] The comparison is of complete pipelines and does not show which ingredient causes the gain.

IG against posterior correlation (RQ2). Holding each fitted GMM fixed, ranking bins by their correlation with the GMM posterior maps outperformed IG in all 48 section–arm comparisons at seed 1 (figure 3). The predeclared rule did not anticipate an outcome this one-sided. Its exact fall-through wording therefore applies: "neither predeclared verdict applies, because IG scored below posterior-|PCC| in every section".

Across seeds. Over training seeds 1–3, correlation outperformed IG in 140 of 144 bin-level comparisons (table III). The four exceptions were small (at most $+0.017$). The strict per-seed rule (IG below in at least 7/8 sections) failed only at CAC seed 2, where IG was below in 6/8 sections, so the predeclared label is "does not hold overall". The direction never reversed at collection level: the mean difference was negative in all 30 dataset × arm × seed × scale cells.

Statistics. With seeds averaged within section, the difference is statistically supported in all six dataset–arm comparisons at

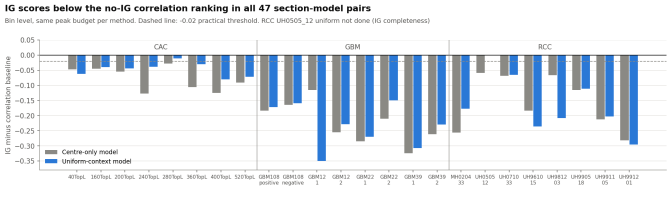


Fig. 3. IG minus posterior correlation (mSCF1) per section, for both VAE arms, at seed 1. Every valid comparison is negative: correlation with the segmentation outperforms IG on the same segmentation.

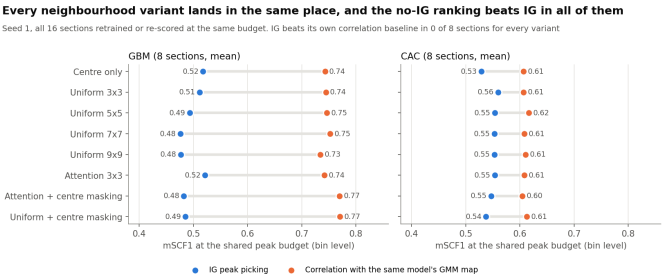


Fig. 4. All neighbourhood variants relative to uniform 3×3 context under IG. No variant gives a consistent, spatially specific gain.

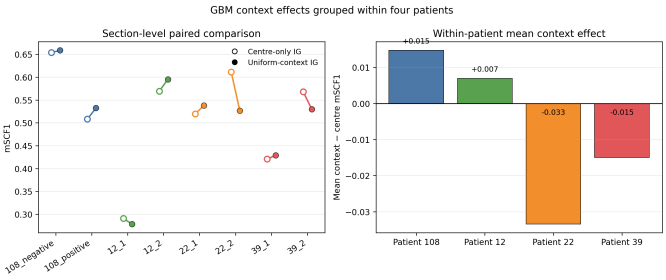


Fig. 5. Uniform minus centre-only IG score on GBM, grouped by PDX line. The direction differs between lines.

bin level (table III). Correlation scored higher in every section (rank-biserial $r = -1$), and every raw p-value is the smallest attainable for $n = 8$, so this is as strong as eight sections can show. At peak level (P1) the result is also supported, except for the CAC uniform arm (-0.054, 7/8, Holm $p = 0.078$). The descriptive mixed models agree and attribute almost no variance to the seed, and GBM is negative in every PDX line. Increasing IG sampling from 12 to 48 or 192 pixels per component raised scores by at most $+0.024$ on the development sections, which did not meet the rule for changing the production setting.

Answer to RQ1 and RQ2. The pipeline beats legacy msiPL, but the attribution method is not what carries that gain. Given the same segmentation, a direct correlation ranking is better than IG on all three datasets. The useful signal is in the label-free segmentation, not in the gradients.

C. Simple rankings and the metric

Among the predeclared label-free rankings, the strongest differs by collection (table II):

- on GBM, posterior correlation (0.743), with Moran's I at 0.684;
- on CAC, mean intensity (0.612), with correlation, Moran's I and variance all within 0.02;
- on RCC, Moran's I (0.63), above correlation (0.57–0.58).

A max-aggregated correlation variant, which was not part of the predeclared comparison, scores 0.626–0.629 on CAC; we report it but draw no conclusion from it.

CAC is close to saturation at its budget. All methods converge near 0.41 at twice the budget, including the oracle. This limits how much CAC differences can reveal. On GBM and RCC, sections whose GMM agrees better with the masks (balanced accuracy) have correlation scores closer to the oracle (Spearman $\rho = -0.90$ and -0.84 against the gap to the oracle). mSCF1 rewards ion images that follow the expert regions. A segmentation that resembles those regions therefore scores well when correlated against, and this alignment is part of why correlation does well (section V).

D. Spatial context in the encoder and the segmentation (RQ3)

- **Uniform 3×3 context.** Under IG, context "helps" on CAC: $+0.023$ (bin) and $+0.027$ (P1) averaged over three seeds, positive in 8/8 sections. It is not statistically supported (Holm $p = 0.13$), and it is 0.000 under correlation ranking. Over three seeds, GBM and RCC are "equivalent" (-0.004, 3/8 and $+0.018$, 6/8). On GBM the direction differs between PDX lines (figure 5).
- **Spatial specificity.** If the CAC gain came from correct local placement, real neighbours should beat shuffled nonlocal ones. They tied: 0.542 for real 3×3 against 0.543 for shuffled. On GBM, zeroed context scored highest (0.459, against 0.446 for real).
- **Larger windows.** 7×7 and 9×9 windows did not meet the "helps" rule. Both were 0.035 below 3×3 on GBM (positive in 2/8, "mixed") and "equivalent" on CAC (figure 4).
- **Learned attention** was "equivalent" to uniform averaging on both collections ($+0.011$ GBM, -0.006 CAC). With real neighbours its weights stayed almost uniform (normalised entropy 0.98–0.999). With shuffled neighbours it learned selective weights, so the mechanism works; real neighbours are simply near-copies of the centre (median cosine about 0.98). Masking the centre during training did not make attention selective, and it lowered GBM scores by 0.039.
- **Spatially smoothed segmentation** was never labelled "helps". Four comparisons were "equivalent" and two "do not meet the 'helps' rule (labelled mixed)": GBM uniform ($+0.029$, driven by two sections) and RCC uniform (-0.021, 4/8).

Answer to RQ3. Neighbourhood information entering the encoder or the segmentation gave no consistent, spatially specific gain in this pipeline. The spatial information that did help sits in the ranking statistic: correlation with spatially coherent segment maps, and Moran's I, which ranks bins by how spatially structured their ion images are.

E. Is the nonlinear encoder needed? (RQ2)

Replacing the VAE latent space with a 5-component PCA, while keeping the GMM and correlation ranking, gave the following VAE − PCA-5 differences:

TABLE II

COLLECTION MEAN MSCF1 AT K_{BIN}, SEED 1, BIN LEVEL. ALL RANKINGS EXCEPT THE BASELINES USE THE SAME FITTED GMM. BOLD: BEST LABEL-FREE RANKING PER ROW. THE ORACLE USES THE EXPERT MASKS AND IS A CEILING.

Collection	VAE arm	IG	Posterior corr.	Moran's I	Mean int.	Variance	Legacy msiPL	Random	Oracle
GBM	centre	0.518	**0.743**	0.684	0.483	0.497	0.414	0.009	0.809
GBM	uniform	0.511	**0.744**	0.684	0.483	0.497	0.414	0.009	0.809
CAC	centre	0.528	0.606	0.605	**0.612**	0.593	0.402	0.108	0.654
CAC	uniform	0.560	0.607	0.605	**0.612**	0.593	0.402	0.108	0.654
RCC	centre	0.412	0.568	**0.631**	0.498	0.508	[CHECK:]	0.171	0.756
RCC	uniform	0.410	0.578	**0.631**	0.498	0.508	[CHECK:]	0.171	0.756

[CHECK: RCC msiPL footnote: scored at its maximum reachable count (item 2)]

TABLE III

IG MINUS POSTERIOR CORRELATION (BIN LEVEL) BY TRAINING SEED. EACH SEED ENTRY IS THE COLLECTION MEAN DIFFERENCE, WITH THE NUMBER OF SECTIONS IN WHICH IG WAS LOWER. MEAN $\pm$ SD IS ACROSS THE THREE SEED-LEVEL MEANS. HL IS THE HODGES–LEHMANN MEDIAN DIFFERENCE OVER SECTIONS (SEEDS AVERAGED WITHIN SECTION) WITH ITS EXACT 95% CI; HOLM ADJUSTS WITHIN THE SIX-TEST RANKING FAMILY.

Collection	Arm	Seed 1	Seed 2	Seed 3	Mean $\pm$ SD	IG lower	HL [95% CI]	Holm p
GBM	centre	-0.225 (8/8)	-0.261 (8/8)	-0.247 (8/8)	-0.244 ± 0.018	24/24	-0.248 [-0.293, -0.186]	0.047
GBM	uniform	-0.233 (8/8)	-0.254 (8/8)	-0.240 (8/8)	-0.242 ± 0.011	24/24	-0.241 [-0.311, -0.181]	0.047
CAC	centre	-0.078 (8/8)	-0.080 (8/8)	-0.074 (7/8)	-0.077 ± 0.003	23/24	-0.078 [-0.116, -0.040]	0.047
CAC	uniform	-0.047 (8/8)	-0.057 (6/8)	-0.060 (8/8)	-0.055 ± 0.007	22/24	-0.046 [-0.080, -0.027]	0.047
RCC	centre	-0.156 (8/8)	-0.129 (7/8)	-0.137 (8/8)	-0.141 ± 0.014	23/24	-0.137 [-0.204, -0.073]	0.047
RCC	uniform	-0.169 (8/8)	-0.134 (8/8)	-0.119 (8/8)	-0.140 ± 0.026	24/24	-0.139 [-0.210, -0.070]	0.047
Total						140/144		

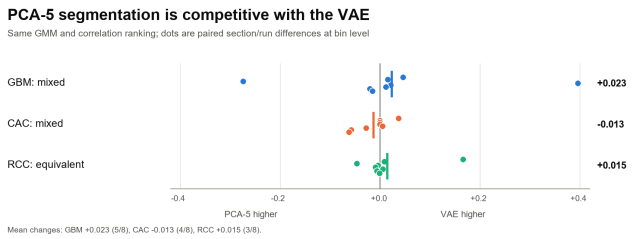


Fig. 6. VAE minus PCA-5 posterior-correlation score per section, using the same GMM and ranking. The verdicts are "mixed" (GBM, CAC) and "equivalent" (RCC).

- GBM: $+0.023$, higher in 5/8 sections, "mixed";
- CAC: -0.013, "mixed";
- RCC: "equivalent" (-0.010).

Averaging the VAE over three seeds did not change the verdicts (figure 6). VAE segmentations matched the masks slightly better (balanced accuracy 0.868 against 0.839 on GBM), without reliably better peaks. Reconstruction did not separate the two either. PCA-5 had *lower* reconstruction error than the VAE on all eight GBM sections, and slightly higher error on 7/8 CAC sections. This matches the finding of Mazarura et al. [2025] that reconstruction quality and downstream usefulness can rank models differently. The number of GMM components was not validated without labels. BIC selected the largest candidate ($K = 6$) in every tested case, including all 16 RCC runs, with no interior minimum, and a finer segmentation lowered GBM scores by about 0.30. [CHECK: training length: VAE scores at 10, 25, 50 and 100 epochs on the development sections (item 3)]

F. S³PL: neighbour dependence, reproduction gap and seeds (RQ3, RQ4)

Neighbour dependence. Retraining S³PL with each patch filled by copies of the centre spectrum collapsed it from 0.591 to 0.18 and 0.15 on CAC (two seeds, 8/8 sections). Retraining with shuffled nonlocal neighbours recovered most of the score (0.55 against 0.59). Correct placement added about $+0.04$ inconsistently (4/8 and 3/8 sections at least $+0.02$), so the label is "mixed" (Holm $p = 0.63$). S³PL needs neighbour input under its patch objective, but a consistent benefit from *spatially correct* neighbours was not established. Training S³PL for longer made it worse: from epoch 10 to 100, CAC fell from 0.591 to 0.335 and GBM from 0.316 to 0.212. Epoch 10 was best in 15 of 16 sections, so the released 10-epoch protocol does not handicap it.

Reproduction gap. With ten released seeds per section (figure 7), S³PL scored 0.319 ± 0.042 on GBM and 0.547 ± 0.038 on CAC (mean $\pm$ SD across seeds of the collection means).

- **GBM:** the published 0.496 is about four SD above our mean and above all ten seed-level means (maximum 0.374). It lies above every one of our seeds in 6/8 sections. Taking the best of ten seeds in every section reaches only 0.434. Labels: "outside our seed distribution"; "not explained by seed selection alone".
- **CAC:** the published 0.601 is just outside our range (maximum 0.591). Best-of-ten reaches 0.5998, which is within the published two-decimal rounding. So CAC is consistent with a favourable seed or best-of-several reporting, but this is not shown.
- **Preprocessing:** replacing the released spatial-maximum normalisation with the TIC normalisation described in the

paper did not raise the GBM mean (0.303 against 0.324; HL -0.012 [-0.154, 0.099]). Label: "does not explain".

The remaining candidate explanations are untested: a different data preparation for specific sections, or hyperparameters tuned on the evaluation sections, which the paper describes.

Why seeds change S^{3}PL. About half of the score variance is between sections, about 0–1% is a seed main effect, and about 47% is a section × seed interaction. Seeds do not shift whole collections; they switch particular sections between distinct outcomes:

- GBM108_positive moves between 0.02 and 0.46;
- CAC 520TopL moves between 0.17 and 0.71;
- in flipping sections, peak lists from different seeds can overlap by as little as 1%.

Separating the seed into its parts on three such sections showed the following:

- weight initialisation dominated data order (SD 0.222 against 0.001 on GBM108_positive, and 0.128 against 0.010 on GBM12_2), giving the label "initialisation dominant";
- on GBM108_negative, data order also mattered (ratio 1.55);
- same-seed reruns were bit-identical, so GPU nondeterminism played no part ("not dominated by nondeterminism").

Under TIC normalisation the per-section seed SD fell from 0.061 to 0.011 and the switching disappeared. Released S^{3}PL therefore reaches at least two distinct solutions on these sections, and its starting weights mostly decide which one.

Answer to RQ4. msiPL reproduces. S^{3}PL reproduces on CAC and RCC but not on GBM, and neither seed selection nor the documented preprocessing explains the GBM gap. S^{3}PL is highly seed-sensitive in specific sections because its initialisation selects between distinct solutions, so single-seed comparisons against it are unreliable.

G. RCC: held-out confirmation and its qualifier

On the held-out RCC collection, correlation outperformed IG in every valid run (centre -0.156 and uniform -0.169, IG higher in 0/8 runs each), and Moran's I was the strongest label-free ranking (0.63). This confirms the main result on a dataset not used for development. One qualifier is necessary. The unannotated class covers 45–78% of pixels, and when only cancer and normal pixels are scored, the advantage shrinks:

- for the centre-only arm, correlation beats IG in 5/8 runs and falls below mean intensity, variance and Moran's I;
- for the uniform arm, correlation beats IG in 6/7 runs [CHECK: UH0505_12 cancer/normal-only rerun].

The three-class RCC result is therefore partly driven by separating annotated tissue from the unannotated region.

H. Runtime

Table IV reports training and ranking times. Once a segmentation exists, correlation takes about a second or less per section, while IG takes seconds to minutes. Every timed ranking reproduced the saved ranking exactly. Cached 100-epoch VAE training takes longer than S^{3}PL's 10-epoch protocol, but remains practical. On CAC section 160TopL, three complete native-workflow repeats (training, selection and scoring) took 122–135 s for the VAE against 151–174 s for S^{3}PL, with peak memory of 1.15–1.18 GiB against 1.31–1.32 GiB (GPU 237 against 315 MiB). These are measurements under each method's native protocol, not an equal-work efficiency comparison.

TABLE IV
MEDIAN RUNTIME PER SECTION. TRAINING: CACHED 100-EPOCH VAE (SEEDS 2–3) AND 10-EPOCH S^{3}PL, ON GPU. RANKING: CPU, TWO THREADS, GIVEN A TRAINED MODEL; IG INCLUDES THE GMM FIT AND FAITHFULNESS CHECKS.

	GBM	CAC	RCC
VAE training (min)	32–35	1.3–1.4	2.2
S^{3}PL training (min)	13.7	0.9	1.6
Posterior corr. (s)	1.12	0.028	0.11
Moran's I (s)	8.4	0.18	0.75
Mean intensity (s)	0.10	0.003	0.013
IG (s)	770	6.0	372

Ranking CPUs: i5-14400 (GBM, CAC) and i9-10940X (RCC, shared node), each matching the CPU that produced the saved rankings.

V. DISCUSSION

A. Why correlation outperforms IG here

Three explanations fit the ranking result.

1) **Different questions.** IG asks which bins would change a few sampled pixels' component probabilities. Correlation asks which ion images follow a component across the whole section, which is closer to what mSCF1 rewards.
2) **Sampling.** IG averages over 12 sampled pixels per component, while correlation uses every pixel. More pixels improved IG only slightly ($\leq +0.024$) on the two sections tested, so sampling explains little of a gap of 0.05–0.25, but other IG settings (baselines, path counts, signed attributions) remain untested.
3) **Metric alignment.** A segmentation that resembles the masks produces posterior maps that a mask-based metric naturally rewards.

Correlation is therefore best read as an ablation finding about this pipeline: the label-free segmentation carries the useful signal, and gradient attribution through a dense VAE does not add to it. We do not claim that IG is generally inferior, nor that segment-then-correlate, which is established practice [Alexandrov and Kobarg 2011], is a new method.

B. What remains of the spatial hypothesis

The original hypothesis was that neighbourhood input would improve a VAE's representation and therefore its peaks. In a dense VAE that averages or attends over neighbours, no intervention gave a consistent gain beyond what nonlocal (shuffled) context also provides, and smoothing the segmentation did not help either. The attention experiments explain why. At 3×3 scale, adjacent spectra are near-copies of the centre, so there is little to select between. Spatial information did help in two other places: as a ranking statistic (Moran's I, and correlation with spatially coherent segments), and in S^{3}PL, which collapses without neighbour input. Even there, though, correctly placed neighbours added only an inconsistent $+0.04$ over shuffled ones. Spatially aware *ranking*, rather than spatially

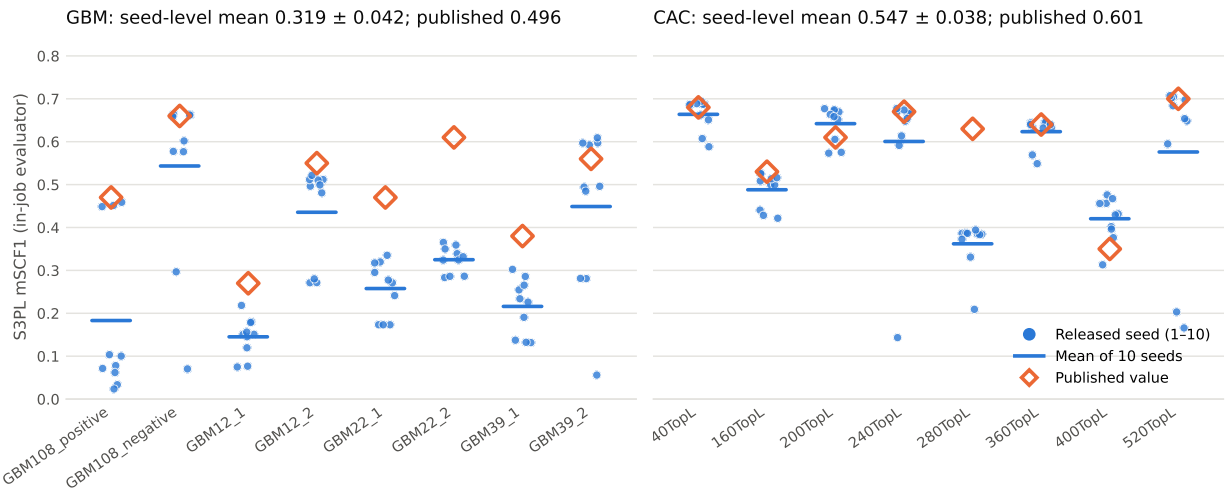


Fig. 7. S^{3}PL mSCF1 per section over ten released seeds (dots), seed mean (bar) and published value (diamond), using S^{3}PL's own evaluator. GBM: the published value is above every seed in 6/8 sections ("outside our seed distribution"). CAC: just outside, within the published rounding. One CAC run (240TopL, seed 7) returned 256 of 257 peaks, as the released code does when fewer distinct bins are selected.

aware *encoding*, is what the evidence supports, and it gives this report its title.

C. Reconstruction is not ranking quality

PCA-5 reconstructed GBM better than the VAE yet ranked peaks about equally well, while the VAE segmented slightly better. Earlier development runs showed the same pattern: context models reconstructed slightly worse but changed which peaks were chosen, and a spatial penalty smoothed the latents while worsening reconstruction. Reconstruction error rewards fitting all intensity variation, including intense ions unrelated to tissue classes, whereas peak quality here depends on how well a low-dimensional segmentation separates the classes. This agrees with Mazarura et al. [2025].

D. S^{3}PL instability and the reproduction gap

The S^{3}PL seed analysis shows that comparisons with a single S^{3}PL run can be misleading. Specific sections switch between distinct solutions depending on the initial weights. The released preprocessing appears to create this instability, since TIC normalisation removes it while also removing the highest-scoring outcomes. Neither seeds nor normalisation explain the published GBM score. We therefore report the gap as unresolved, and we do not select favourable seeds or checkpoints for any baseline.

E. Label-free operation and cost

The pipeline fits and ranks without labels, but two settings still come from annotation: the number of segments and the peak budget. BIC could not replace the first, so the pipeline is not yet autonomous. On cost, the dense VAE used comparable time and less memory than S^{3}PL on the section measured, and once a segmentation exists, correlation ranking takes about a second where IG takes minutes. The lightweight model is therefore practical, but it did not gain from its spatial input, which was the reason for adding it.

F. Limitations and threats to validity

- **Statistical units.** GBM has four PDX lines, CAC one biopsy and RCC eight subjects. P-values describe these sections, not a population. With $n = 8$, only small families can reach Holm significance, so many comparisons rest on practical labels and effect sizes.
- **Class-informed configuration.** The number of GMM components equals the number of annotated classes and was not supported by BIC. Peak budgets come from published, annotation-related counts. The pipeline is label-free in fitting and ranking only.
- **The metric.** mSCF1 measures agreement with expert masks, not molecular identity. Methods that produce mask-like maps are favoured.
- **RCC.** The three-class result partly reflects separating annotated tissue from a large unannotated region.
- **Post hoc elements.** The PCA control, the smoothed segmentation, the seed replication and the S^{3}PL seed analysis were chosen after earlier results were known. Each was predeclared before its own outputs, and all are reported, including null and negative outcomes. Recommending correlation is a post hoc conclusion.
- **Scope.** The VAE uses fixed-window, dense neighbourhood input. Graph-based and ion-image-level spatial models, cross-section transfer, and other attribution settings were not tested.

VI. Conclusion

This report tested whether local spatial context helps a lightweight VAE select informative MSI peaks, and which part of a learned pipeline determines peak quality. Under matched budgets and predeclared rules, on three public collections, we reached the following conclusions.

- Ranking peaks by correlation with a label-free GMM segmentation outperformed gradient attribution through the VAE on every dataset. This was statistically supported and stable across training seeds.

- Neighbourhood input to the encoder, and smoothing of the segmentation, gave no consistent, spatially specific gain. Spatial information helped as a ranking statistic, through segmentation correlation and Moran's I.
- A linear PCA segmentation performed comparably to the VAE, and reconstruction error did not track peak quality.
- msiPL reproduced. S^3PL's published GBM score did not, and seeds altered S^3PL by switching sections between distinct solutions, mainly through weight initialisation.

The contribution is a controlled evaluation, not a new peak picker. It shows where spatial information pays off in learned MSI peak ranking, and how strongly conclusions depend on fair budgets, model-free references and seed replication.

Future work should first:

- validate a label-free choice of the number of segments, which BIC could not provide;
- test a predeclared combination of segmentation correlation and Moran's I on new data;
- evaluate spatial models that operate on graphs or whole ion images rather than fixed windows;
- test whether a model fitted on one section transfers to another.

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